

Mark Scheme — Practice Paper 4 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 Term = $\binom{8}{2}(2)^6 \left(-\frac{x}{2}\right)^2$ (M1) = $28 \times 64 \times \frac{1}{4}x^2$ (A1) $\Rightarrow$ coefficient 448 (A1). [3]

2 $2 \sin \theta \cdot \frac{\sin \theta}{\cos \theta} = 3 \Rightarrow 2 \sin^2 \theta = 3 \cos \theta$ (M1). $2(1 - \cos^2 \theta) = 3 \cos \theta \Rightarrow 2 \cos^2 \theta + 3 \cos \theta - 2 = 0$ (A1). $(2 \cos \theta - 1)(\cos \theta + 2) = 0 \Rightarrow \cos \theta = \frac{1}{2}$ (M1; $\cos \theta = -2$ rejected). $\theta = 60^\circ, 300^\circ$ (A1). [4]

3(a) $60 + 19d = 41$ (M1) $\Rightarrow d = -1$, i.e. each rung is 1 cm shorter (A1). [2]

3(b) $S_{20} = \frac{20}{2}(60 + 41)$ (M1) = 1010 cm (A1). [2]

4(a) Gradient = $\frac{-1-7}{6-2} = -2$ (B1). $y - 7 = -2(x - 2) \Rightarrow y = -2x + 11$ (B1). [2]

4(b) Method 1: $C(c, 0)$ with $AC^2 = BC^2$: $(c - 2)^2 + 49 = (c - 6)^2 + 1$ (M1) $\Rightarrow -4c + 53 = -12c + 37$ (A1) $\Rightarrow c = -2$, $C(-2, 0)$ (A1). Method 2: perpendicular bisector: midpoint $(4, 3)$, gradient $\frac{1}{2}$: $y - 3 = \frac{1}{2}(x - 4)$ (M1 A1); $y = 0 \Rightarrow x = -2$ (A1). [3]

5(a) Arc $AB = 10 \times 0.8 = 8$, arc $CD = 6 \times 0.8 = 4.8$ (B1 both). $AC = BD = 4$ (B1). Perimeter = $8 + 4.8 + 4 + 4 = 20.8$ cm (B1). [3]

5(b) Area = $\frac{1}{2}(10)^2(0.8) - \frac{1}{2}(6)^2(0.8)$ (M1 A1) = $40 - 14.4 = 25.6$ cm² (A1). [3]

6(a) $gf(x) = (3x - 2)^2 + 4(3x - 2)$ (M1) = $9x^2 - 12x + 4 + 12x - 8 = 9x^2 - 4$ (A1). [2]

6(b) $9x^2 - 4 = 32 \Rightarrow x^2 = 4$ (M1) $\Rightarrow x = \pm 2$ (A1; both needed). [2]

6(c) g is not one-one (e.g. $g(0) = g(-4) = 0$; or "the graph fails the horizontal line test") (B1). $g(x) = (x + 2)^2 - 4$, so least $k = -2$ (B1). [2]

7(a) $x^2 - 4x + 6 = mx + 2 \Rightarrow x^2 - (4 + m)x + 4 = 0$ (M1 A1). Tangent $\Rightarrow (4 + m)^2 - 16 = 0$ (M1) $\Rightarrow 4 + m = \pm 4 \Rightarrow m = 0$ or $m = -8$ (A1). [4]

7(b) $m = 0$: $x^2 - 4x + 4 = 0 \Rightarrow x = 2$, point $(2, 2)$ (M1 A1). $m = -8$: $x^2 + 4x + 4 = 0 \Rightarrow x = -2$, $y = -8(-2) + 2 = 18$, point $(-2, 18)$ (A1). [3]

8(a) $\frac{dy}{dx} = 3x^2 - 6x - 9$ (B1). Decreasing $\Rightarrow 3x^2 - 6x - 9 < 0 \Rightarrow 3(x - 3)(x + 1) < 0$ (M1 A1). $-1 < x < 3$ (A1). [4]

8(b) At $x = 0$: $y = 5$, $\frac{dy}{dx} = -9$ (B1). Normal gradient $\frac{1}{9}$ (M1). $y - 5 = \frac{1}{9}x \Rightarrow x - 9y + 45 = 0$ (A1). [3]

9(a) $y = \int (3x^{1/2} - x^{-2}) dx = 2x^{3/2} + x^{-1} + c$ (M1 A1 A1). $(1, 5)$: $2 + 1 + c = 5 \Rightarrow c = 2$: $y = 2x^{3/2} + \frac{1}{x} + 2$ (A1). [4]

9(b) Curve meets $y = 5$ at $x = 2$ (B1). Area = $\int_0^2 (5 - (x^2 + 1)) dx = \int_0^2 (4 - x^2) dx$ (M1) = $\left[4x - \frac{1}{3}x^3\right]_0^2 = 8 - \frac{8}{3}$ (A1) = $\frac{16}{3}$ (A1). [Alternative $\int_1^5 \sqrt{y-1} dy = \frac{2}{3}(4)^{3/2} = \frac{16}{3}$.] [4]

10(a) $ar = 6$, $ar^3 = 54 \Rightarrow r^2 = 9$ (M1) $\Rightarrow r = 3$ (positive terms), $a = 2$ (A1). $S_8 = \frac{2(3^8 - 1)}{3 - 1}$ (M1) = 6560 (A1). [4]

10(b) Terms $a, a + 4, a + 12$ (B1). $(a + 4)^2 = a(a + 12)$ (M1) $\Rightarrow a^2 + 8a + 16 = a^2 + 12a \Rightarrow a = 4$ (A1). GP: 4, 8, 16, so $r = 2$ (A1). **[4]**

11(a) $(1 - \cos \theta) \cdot \frac{\cos \theta + 1}{\cos \theta}$ (M1) $= \frac{(1 - \cos \theta)(1 + \cos \theta)}{\cos \theta} = \frac{1 - \cos^2 \theta}{\cos \theta}$ (M1) $= \frac{\sin^2 \theta}{\cos \theta}$ (A1 AG). **[3]**

11(b) $\frac{\sin^2 \theta}{\cos \theta} = \frac{3}{2} \sin \theta \Rightarrow \sin \theta \left(\frac{\sin \theta}{\cos \theta} - \frac{3}{2} \right) = 0$ (M1). $\sin \theta = 0 \Rightarrow \theta = 180^\circ$ (A1; $0^\circ, 360^\circ$ excluded).

$\tan \theta = \frac{3}{2} \Rightarrow \theta = 56.3^\circ$ (A1), 236.3° (A1). **[4]**

11(c) $\sin \left(x + \frac{\pi}{6} \right) = \frac{1}{2}$, with $x + \frac{\pi}{6} \in \left[\frac{\pi}{6}, \frac{13\pi}{6} \right]$ (M1). $x + \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$ (A1) $\Rightarrow x = 0, \frac{2\pi}{3}, 2\pi$ (A1; all three needed). **[3]**

12(a) $(x - 2)^2 + (y + 3)^2 = 25$: centre $(2, -3)$ (B1), radius 5 (B1). **[2]**

12(b) $y = 0: x^2 - 4x - 12 = 0$ (M1) $\Rightarrow (x - 6)(x + 2) = 0$ (A1) $\Rightarrow (6, 0)$ and $(-2, 0)$ (A1). **[3]**

12(c) $y = 1: x^2 - 4x - 5 = 0 \Rightarrow x = 5, -1$ (M1). $PQ = 6$ (A1). **[2]**

Total: 75